

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr G .A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Scenario-Based Assessment

DURATION: 1.5 Hrs

6. A financial trading system applies Reinforcement Learning to decide buy/sell actions based on market signals. Analyze how states, actions, and rewards can be defined in this context and evaluate the risks of delayed rewards in volatile markets.

A. Scenario-Based Analysis: Reinforcement Learning for Financial Trading

Consider an AI-based financial trading system that uses Reinforcement Learning (RL) to decide whether to Buy, Sell, or Hold a particular financial asset based on changing market conditions.

At each time step, the RL agent observes the current market situation, selects an action, and receives a reward based on the outcome of that decision. Its goal is to learn a trading policy that maximizes long-term returns while controlling risk.

1. State Definition

The **state** represents the information available to the RL agent when making a trading decision.

A possible state can include:

[
 $S_t = [P_t, R_t, MA_t, V_t, RSI_t, Position_t, Balance_t]$
]

Where:

State Component	Description
Current Price	Latest market price of the asset
Historical Prices	Previous price movements
Returns	Recent percentage changes
Moving Average	Indicates market trend
Trading Volume	Indicates market activity
RSI / Technical Indicators	Indicates momentum or possible overbought/oversold conditions
Current Position	Whether the agent currently holds the asset
Available Balance	Capital available for trading
Portfolio Value	Current total value of holdings and cash
Volatility	Level of market price fluctuation

Example Scenario

Suppose the market state is:

Current Price = ₹500

Moving Average = ₹480

RSI = 40

Trading Volume = High

Current Position = No Holdings

Market Volatility = Medium

The RL agent uses this information to determine the most appropriate action.

2. Action Definition

The **action space** represents the decisions available to the trading agent.

A simple discrete action space is:

Action	Meaning
0	Hold
1	Buy
2	Sell

Thus:

[
A = {Hold, Buy, Sell}
]

A more advanced system may allow:

- Buy 10% of available capital
- Buy 25% of available capital
- Sell 25% of holdings
- Sell all holdings
- Hold the current position

This creates a larger action space and allows the agent to learn both **trading direction** and **position size**.

Example Decision

If the agent observes a possible upward trend:

State:

Price is increasing

RSI = 40

Moving average indicates upward trend

No current position

Action:

BUY

If the asset is already owned and the market begins to decline:

State:

Price decreasing rapidly

High volatility

Current position = Holding

Action:

SELL

3. Reward Definition

The reward should represent the quality of a trading decision.

A simple reward can be based on the change in portfolio value:

$$[R_t = PortfolioValue_{\{t+1\}} - PortfolioValue_t]$$

For example:

Profitable Buy

Buy Price = ₹500

Future Price = ₹550

Reward = +₹50

The agent receives a positive reward.

Unprofitable Buy

Buy Price = ₹500

Future Price = ₹450

Reward = -₹50

The agent receives a negative reward.

Risk-Aware Reward Function

A more realistic reward can include transaction costs and risk penalties:

$$R_t = \text{Profit}_t - \lambda(\text{TransactionCost}_t + \text{Risk}_t)$$

Where:

- (Profit_t) = Change in portfolio value
- $(\text{TransactionCost}_t)$ = Brokerage and transaction costs
- (Risk_t) = Market or portfolio risk
- (λ) = Risk penalty weight

This prevents the agent from learning a strategy that earns high returns but takes excessive risks.

4. Scenario of Delayed Rewards

A major challenge in financial RL is that the result of an action may not be immediately visible.

Example

At time (t_1) :

Market Price = ₹500

Agent Action = BUY

Immediately after buying, the market may decrease:

Time t_2 :

Price = ₹470

Immediate Result = Loss

However, after several days:

Time t_3 :

Price = ₹600

Final Result = Profit

Therefore, the initial buy action may appear to be incorrect when evaluated immediately, even though it eventually produces a positive return.

This is known as the **delayed reward problem**.

5. Risks of Delayed Rewards in Volatile Markets

Risk 1: Difficulty in Identifying Good Actions

In a volatile market, prices can change rapidly.

A good action may initially produce a loss:

BUY → Temporary Price Drop → Later Price Increase

The RL agent may incorrectly learn that the buying action was bad.

Risk 2: Incorrect Credit Assignment

The agent may perform several actions before receiving a significant reward.

For example:

BUY

↓

HOLD

↓

HOLD

↓

SELL

↓

Profit

The RL algorithm must determine which action contributed most to the final profit.

This is called the credit assignment problem.

It may be difficult to decide whether the profit resulted from:

- The original buy decision
- The holding strategy
- The timing of the sell decision
- A sudden external market movement

Risk 3: High Market Volatility

In volatile markets:

[
Price_{t+1} \neq Predictable(Price_t)
]

Unexpected events can rapidly change market conditions.

For example:

Agent decides: BUY

↓

Unexpected negative news

↓

Market crashes

↓

Large negative reward

The agent may be penalized even if its decision was reasonable based on the available market signals.

Risk 4: Sparse Rewards

If rewards are given only after a complete trading cycle, such as:

BUY → HOLD → HOLD → SELL → Reward

the agent receives very little feedback during learning.

This can result in:

- Slow learning

- Poor exploration
- Difficulty learning long-term strategies

Risk 5: Overreaction to Short-Term Market Changes

If the reward focuses too strongly on immediate profit, the agent may learn:

Small price decrease → SELL immediately

Small price increase → BUY immediately

This may lead to excessive trading.

Consequences include:

- Higher transaction costs
- Reduced long-term returns
- Unstable trading behavior

6. Scenario-Based Comparison

Scenario	State	Action	Immediate Result	Delayed Result	RL Challenge
Upward Trend	Price rising, positive indicators	Buy	Small loss	Large profit	Delayed reward
Sudden Crash	Positive signals followed by bad news	Buy	Loss	Large loss	Market uncertainty
Temporary Decline	Price drops temporarily	Hold	Negative reward	Later profit	Correct credit assignment
High Volatility	Rapid price changes	Hold	Small opportunity loss	Reduced risk	Balancing profit and safety

Profit Taking	Price reaches target	Sell	Immediate profit	Possible future missed gains	Timing decisions
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7. How RL Can Handle Delayed Rewards

Several techniques can help reduce the impact of delayed rewards.

Discount Factor

The discount factor (γ) allows the agent to consider future rewards:

$$G_t = R_t + \gamma R_{t+1} + \gamma^2 R_{t+2} + \dots$$

A higher value of (γ) gives more importance to long-term rewards.

For financial trading, a carefully selected value helps the agent balance:

- Immediate profit
- Future portfolio growth

Experience Replay

Past trading experiences can be stored and reused:

(State, Action, Reward, Next State)

This can improve learning efficiency and reduce dependence on only recent market events.

Reward Shaping

Instead of giving rewards only at the end of a trade, the system can provide intermediate feedback based on:

- Portfolio value changes
- Risk reduction
- Drawdown
- Transaction costs

- Volatility

For example:

[
Reward =
Portfolio\ Return

Risk\ Penalty

Transaction\ Cost
]

Risk-Sensitive Reinforcement Learning

The agent should not optimize only for profit.

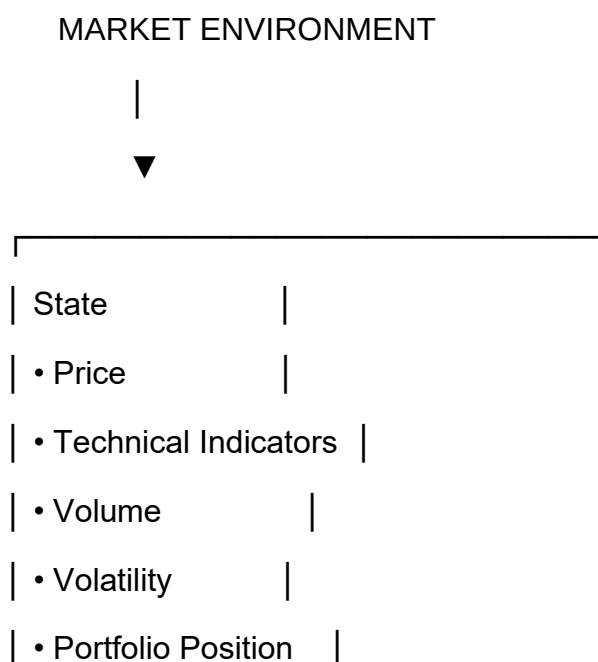
Additional risk metrics can be included, such as:

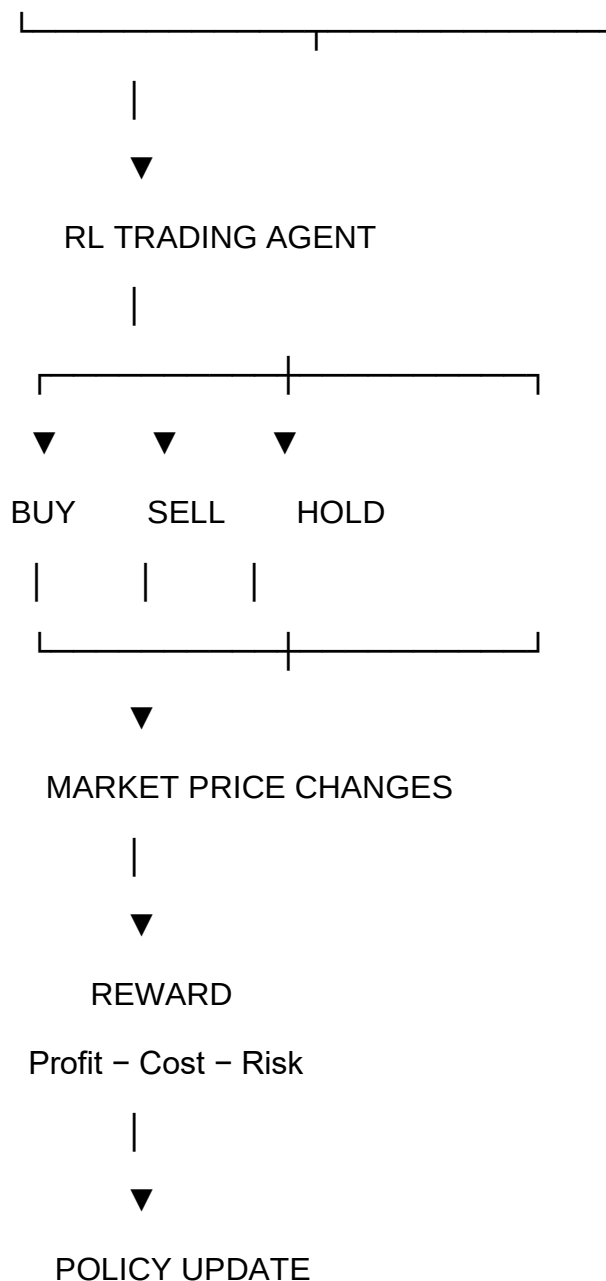
- Maximum drawdown
- Portfolio volatility
- Value at Risk
- Sharpe ratio or similar risk-adjusted performance measures

This helps reduce the possibility of a highly profitable but extremely risky strategy.

8. Overall Analysis

The RL trading system can be represented as:





9. Conclusion

In this scenario, the state represents market signals, asset information, and portfolio conditions. The actions represent trading decisions such as Buy, Sell, and Hold. The reward can be defined using portfolio profit while also considering transaction costs and financial risk.

The major challenge is delayed rewards, particularly in volatile markets. An action that appears unsuccessful in the short term may become profitable later, while external events can create rewards or losses unrelated to the quality of the agent's decision. Therefore, a financial RL system should use appropriate reward design, long-term evaluation, risk penalties, and robust validation rather than optimizing only for immediate profit. **(5 Marks)**

Code of Conduct:

I K.Mounith(192425090)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *K.mounith*

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand